

Math 54 Discussion Worksheet

Midterm Review: Sections 4.1–4.6, 5.1–5.5, 6.1–6.5

Name: _____ ID: _____

Problems

- 1. (4.1 Vector Spaces)**
- Let

$$V = \left\{ \begin{pmatrix} a-b \\ 2a+b \\ 3a \end{pmatrix} : a, b \in \mathbb{R} \right\} \subseteq \mathbb{R}^3.$$

Prove that V is a vector space and determine whether

$$\mathbf{u} = \begin{pmatrix} 1 \\ 5 \\ 6 \end{pmatrix}$$

belongs to V .

- 2. (4.2 Subspaces of $\mathbb{R}^n$)**
- Let

$$W = \{(x, y, z, w) \in \mathbb{R}^4 : x + 2y - z = 0, \quad 3x + 6y - 3z = 0\}.$$

Find a basis for W and its dimension.

- 3. (4.3 Linear Independence)**
- Determine whether the polynomials

$$p_1(t) = 1 + t, \quad p_2(t) = 1 - t, \quad p_3(t) = 1 + t^2$$

are linearly independent in P_2 .

- 4. (4.4 Basis and Dimension)**
- Find a basis for the subspace of
- $\mathbb{R}^4$
- defined by

$$x_1 + x_2 + x_3 + x_4 = 0, \quad x_1 + 2x_2 + 3x_3 + 4x_4 = 0.$$

Then find its dimension.

- 5. (4.5 Row Space, Column Space, and Null Space)**
- Let

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{pmatrix}.$$

Find a basis for $\text{Col}(A)$, a basis for $\text{Row}(A)$, and a basis for $\text{Null}(A)$.

6. (4.6 Coordinate Systems / Change of Basis) Let

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

Show that $\mathcal{B}$ is a basis for $\mathbb{R}^3$, and find $[\mathbf{x}]_{\mathcal{B}}$ for

$$\mathbf{x} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}.$$

7. (5.1 Eigenvalues and Eigenvectors) For

$$A = \begin{pmatrix} 4 & 1 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -2 \end{pmatrix},$$

find all eigenvalues and a basis for each eigenspace.

8. (5.2 The Characteristic Equation) Find the characteristic polynomial and the eigenvalues of

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

9. (5.3 Diagonalization) Let

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}, \quad A = PDP^{-1}.$$

Compute A^{10} explicitly.

10. (5.4 Eigenvectors) For

$$A = \begin{pmatrix} 5 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix},$$

find all eigenvectors corresponding to each eigenvalue.

11. (5.5 Complex Eigenvalues) Let

$$A = \begin{pmatrix} 1 & -2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Find the characteristic polynomial of A , all eigenvalues over $\mathbb{C}$, and a basis for the eigenspace corresponding to $\lambda = 1 + 2i$.

12. (6.1 Inner Product, Length, and Orthogonality) In $\mathbb{R}^3$, let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}.$$

Compute $\mathbf{u} \cdot \mathbf{v}$, $\|\mathbf{u}\|$, $\|\mathbf{v}\|$, and the angle between them.

13. (6.2 Orthogonal Sets) Determine whether the set

$$\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix} \right\}$$

is an orthogonal basis for $\mathbb{R}^4$.

14. (6.3 Orthogonal Projections) Let

$$\mathbf{y} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Find $\text{proj}_{\mathbf{u}}\mathbf{y}$ and decompose $\mathbf{y}$ as

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

where $\hat{\mathbf{y}} \in \text{Span}\{\mathbf{u}\}$ and $\mathbf{z} \perp \mathbf{u}$.

15. (6.4 Orthogonal Bases) Let

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}.$$

Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal basis for $\mathbb{R}^3$, and write

$$\mathbf{x} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$$

as a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$.

16. (6.5 Least-Squares Problems) Find the least-squares solution to

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 4 \end{pmatrix}.$$

Then write down the corresponding least-squares line $y = c + dx$.

Solutions

1. Since

$$\begin{pmatrix} a-b \\ 2a+b \\ 3a \end{pmatrix} = a \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + b \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix},$$

we have

$$V = \text{Span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \right\},$$

so V is a subspace, hence a vector space. To check whether $\mathbf{u} \in V$, solve

$$a - b = 1, \quad 2a + b = 5, \quad 3a = 6.$$

From $3a = 6$, $a = 2$. Then $a - b = 1$ gives $b = 1$, and $2a + b = 5$ checks. Hence

$$\mathbf{u} \in V.$$

2. The second equation is a multiple of the first, so

$$W = \{(x, y, z, w) : x + 2y - z = 0\}.$$

Let $y = s$, $z = t$, $w = u$. Then $x = -2s + t$, so

$$(x, y, z, w) = s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + u \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

A basis is

$$\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}, \quad \dim W = 3.$$

3. Suppose

$$c_1(1+t) + c_2(1-t) + c_3(1+t^2) = 0.$$

Then

$$(c_1 + c_2 + c_3) + (c_1 - c_2)t + c_3t^2 = 0.$$

Matching coefficients:

$$c_1 + c_2 + c_3 = 0, \quad c_1 - c_2 = 0, \quad c_3 = 0.$$

So $c_3 = 0$, $c_1 = c_2$, and then $2c_1 = 0$, hence $c_1 = c_2 = 0$. Therefore the polynomials are linearly independent.

4. From

$$x_1 + x_2 + x_3 + x_4 = 0, \quad x_1 + 2x_2 + 3x_3 + 4x_4 = 0,$$

subtract the first from the second:

$$x_2 + 2x_3 + 3x_4 = 0.$$

So $x_2 = -2x_3 - 3x_4$, and then

$$x_1 = -(x_2 + x_3 + x_4) = x_3 + 2x_4.$$

Let $x_3 = s$, $x_4 = t$. Then

$$\mathbf{x} = s \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 2 \\ -3 \\ 0 \\ 1 \end{pmatrix}.$$

A basis is

$$\left\{ \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -3 \\ 0 \\ 1 \end{pmatrix} \right\}, \quad \dim = 2.$$

5. Row-reduce:

$$\begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Pivot columns are 1 and 3, so a basis for $\text{Col}(A)$ is given by the corresponding original columns:

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

A basis for $\text{Row}(A)$ is the set of nonzero rows of the echelon form:

$$\{(1, 2, 1, 0), (0, 0, -2, 1)\}.$$

For $\text{Null}(A)$, solve

$$x_1 + 2x_2 + x_3 = 0, \quad -2x_3 + x_4 = 0.$$

Let $x_2 = s$, $x_3 = t$. Then $x_4 = 2t$ and $x_1 = -2s - t$. Hence

$$\mathbf{x} = s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \\ 2 \end{pmatrix}.$$

So a basis for $\text{Null}(A)$ is

$$\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\}.$$

6. Let

$$c_1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Then

$$c_1 + c_2 = 0, \quad c_1 + c_3 = 0, \quad c_2 + c_3 = 0.$$

This implies $c_1 = c_2 = c_3 = 0$, so the vectors are linearly independent. Since there are 3 vectors in $\mathbb{R}^3$, $\mathcal{B}$ is a basis.

To find coordinates, solve

$$c_1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}.$$

Thus

$$c_1 + c_2 = 4, \quad c_1 + c_3 = 5, \quad c_2 + c_3 = 6.$$

From these,

$$c_1 = \frac{3}{2}, \quad c_2 = \frac{5}{2}, \quad c_3 = \frac{7}{2}.$$

Therefore

$$[\mathbf{x}]_{\mathcal{B}} = \begin{pmatrix} 3/2 \\ 5/2 \\ 7/2 \end{pmatrix}.$$

7. Since A is upper triangular, the eigenvalues are the diagonal entries:

$$\lambda = 4 \text{ (alg. mult. 2)}, \quad \lambda = -2.$$

For $\lambda = 4$:

$$A - 4I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -6 \end{pmatrix}.$$

So $x_2 = 0$, $x_3 = 0$, and x_1 is free. Thus

$$E_4 = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}.$$

For $\lambda = -2$:

$$A + 2I = \begin{pmatrix} 6 & 1 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

So $x_2 = 0$, $x_1 = 0$, and x_3 is free. Thus

$$E_{-2} = \text{Span} \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

8.

$$A - \lambda I = \begin{pmatrix} 2 - \lambda & 1 & 0 \\ 1 & 2 - \lambda & 1 \\ 0 & 1 & 2 - \lambda \end{pmatrix}.$$

Compute

$$p(\lambda) = \det(A - \lambda I) = (2 - \lambda)((2 - \lambda)^2 - 1) - 1(2 - \lambda).$$

So

$$p(\lambda) = (2 - \lambda)((2 - \lambda)^2 - 2).$$

Equivalently,

$$p(\lambda) = (2 - \lambda)(\lambda^2 - 4\lambda + 2).$$

The characteristic equation is

$$(2 - \lambda)((2 - \lambda)^2 - 2) = 0.$$

Hence the eigenvalues are

$$\lambda = 2, \quad \lambda = 2 + \sqrt{2}, \quad \lambda = 2 - \sqrt{2}.$$

9. Since

$$A = PDP^{-1},$$

we have

$$A^{10} = PD^{10}P^{-1}.$$

Now

$$D^{10} = \begin{pmatrix} 4^{10} & 0 \\ 0 & 1 \end{pmatrix}.$$

Also,

$$P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Thus

$$A^{10} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 4^{10} & 0 \\ 0 & 1 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Carrying this out,

$$A^{10} = \frac{1}{2} \begin{pmatrix} 4^{10} + 1 & 4^{10} - 1 \\ 4^{10} - 1 & 4^{10} + 1 \end{pmatrix}.$$

10. Since A is upper triangular, eigenvalues are 5 and 3.

For $\lambda = 5$:

$$A - 5I = \begin{pmatrix} 0 & 2 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

So $x_2 = 0$, $x_3 = 0$, x_1 free. Thus eigenvectors are all nonzero multiples of

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

For $\lambda = 3$:

$$A - 3I = \begin{pmatrix} 2 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

So $x_1 = -x_2$, with x_2, x_3 free. Thus the eigenspace is

$$E_3 = \left\{ \begin{pmatrix} -s \\ s \\ t \end{pmatrix} : (s, t) \neq (0, 0) \right\} = \text{Span} \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

11. Since A is block diagonal with a 2×2 block and the scalar 3, we have

$$p(\lambda) = \det \begin{pmatrix} 1-\lambda & -2 & 0 \\ 2 & 1-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{pmatrix} = (3-\lambda) \det \begin{pmatrix} 1-\lambda & -2 \\ 2 & 1-\lambda \end{pmatrix}.$$

Now

$$\det \begin{pmatrix} 1-\lambda & -2 \\ 2 & 1-\lambda \end{pmatrix} = (1-\lambda)^2 + 4.$$

So the characteristic polynomial is

$$p(\lambda) = (3-\lambda)((1-\lambda)^2 + 4).$$

Hence the eigenvalues over $\mathbb{C}$ are

$$\lambda = 3, \quad \lambda = 1 + 2i, \quad \lambda = 1 - 2i.$$

For $\lambda = 1 + 2i$,

$$A - (1 + 2i)I = \begin{pmatrix} -2i & -2 & 0 \\ 2 & -2i & 0 \\ 0 & 0 & 2 - 2i \end{pmatrix}.$$

The third equation gives $x_3 = 0$. The first equation is

$$-2ix_1 - 2x_2 = 0 \implies ix_1 + x_2 = 0 \implies x_2 = -ix_1.$$

So taking $x_1 = 1$, we get an eigenvector

$$\begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}.$$

Therefore a basis for the eigenspace corresponding to $\lambda = 1 + 2i$ is

$$\left\{ \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix} \right\}.$$

12.

$$\mathbf{u} \cdot \mathbf{v} = 1 \cdot 2 + 2(-1) + 2 \cdot 2 = 4.$$

$$\|\mathbf{u}\| = \sqrt{1 + 4 + 4} = 3, \quad \|\mathbf{v}\| = \sqrt{4 + 1 + 4} = 3.$$

So

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{4}{9}.$$

Hence

$$\theta = \cos^{-1} \left(\frac{4}{9} \right).$$

13. Check pairwise dot products:

$$(1, 1, 0, 0) \cdot (1, -1, 0, 0) = 0,$$

$$(0, 0, 1, 1) \cdot (0, 0, 1, -1) = 0,$$

and vectors from the first pair are orthogonal to vectors from the second pair because their supports are disjoint. So the set is orthogonal. None of the vectors is zero. There are 4 orthogonal nonzero vectors in $\mathbb{R}^4$, so they form a basis. Hence it is an orthogonal basis for $\mathbb{R}^4$.

14.

$$\text{proj}_{\mathbf{u}} \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}.$$

Now

$$\mathbf{y} \cdot \mathbf{u} = 3 + 1 + 0 = 4, \quad \mathbf{u} \cdot \mathbf{u} = 1 + 1 = 2.$$

So

$$\text{proj}_{\mathbf{u}} \mathbf{y} = 2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}.$$

Thus

$$\hat{\mathbf{y}} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}, \quad \mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

Indeed,

$$\mathbf{z} \cdot \mathbf{u} = 1 - 1 + 0 = 0.$$

15. Compute:

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = 1 - 1 + 0 = 0, \quad \mathbf{u}_1 \cdot \mathbf{u}_3 = 1 + 1 - 2 = 0, \quad \mathbf{u}_2 \cdot \mathbf{u}_3 = 1 - 1 + 0 = 0.$$

So the set is orthogonal; since all are nonzero, it is linearly independent. With 3 vectors in $\mathbb{R}^3$, it is an orthogonal basis.

For an orthogonal basis,

$$\mathbf{x} = c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + c_3 \mathbf{u}_3, \quad c_i = \frac{\mathbf{x} \cdot \mathbf{u}_i}{\mathbf{u}_i \cdot \mathbf{u}_i}.$$

Compute:

$$\mathbf{x} \cdot \mathbf{u}_1 = 2 + 3 + 4 = 9, \quad \mathbf{u}_1 \cdot \mathbf{u}_1 = 3,$$

so $c_1 = 3$.

$$\mathbf{x} \cdot \mathbf{u}_2 = 2 - 3 + 0 = -1, \quad \mathbf{u}_2 \cdot \mathbf{u}_2 = 2,$$

so $c_2 = -\frac{1}{2}$.

$$\mathbf{x} \cdot \mathbf{u}_3 = 2 + 3 - 8 = -3, \quad \mathbf{u}_3 \cdot \mathbf{u}_3 = 6,$$

so $c_3 = -\frac{1}{2}$.

Therefore

$$\mathbf{x} = 3\mathbf{u}_1 - \frac{1}{2}\mathbf{u}_2 - \frac{1}{2}\mathbf{u}_3.$$

16. Let

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 4 \end{pmatrix}.$$

The normal equations are

$$A^T A \mathbf{x} = A^T \mathbf{b}.$$

Compute

$$A^T A = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}, \quad A^T \mathbf{b} = \begin{pmatrix} 9 \\ 18 \end{pmatrix}.$$

So solve

$$\begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 9 \\ 18 \end{pmatrix}.$$

From the first equation,

$$4c + 6d = 9.$$

From the second,

$$6c + 14d = 18.$$

Solving gives

$$d = \frac{3}{2}, \quad c = 0.$$

Thus the least-squares solution is

$$\begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{3}{2} \end{pmatrix},$$

and the least-squares line is

$$y = \frac{3}{2}x.$$